Recap Checkpoint 6 — Full Year 12 Cumulative (Pre-Mock): 1.1, 1.3–1.5 + 2.1–2.3 — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Q1 [3] (AO1). Any three (1 each, factor + effect): - **Clock speed** — more cycles per second → more instructions executed per second. - **Number of cores** — more cores allow more instructions/tasks to run in **parallel**. - **Cache size/amount** — larger/faster cache reduces fetches from slower main memory, speeding access. - (Also accept word/bus width, pipelining.)

Examiner tip: each mark needs a named factor AND its effect — a bare list of words scores less.

Q2 [4] (AO2). - (a) 181 = `1011 0101` (1). *(128+32+16+4+1 = 181)* - (b) 181 ÷ 16 = 11 remainder 5 → high nibble 11 = B, low nibble 5 = 5 → **B5** (1 method, 1 answer). *(Check: `1011`=B, `0101`=5.)* - (c) `1010 0110` = 128+32+4+2 = **166** (1).

Examiner tip: nibble-check binary↔hex: 181 = `1011 0101` = B5 — the two must agree.

Q3 [4] (AO2). −40 + 25 in 8-bit two's complement: - (a) 40 = `0010 1000`, 25 = `0001 1001` (1 for both). - (b) −40: invert `0010 1000` → `1101 0111`, add 1 → **`1101 1000`** (1). - (c) `1101 1000` + `0001 1001` = `1111 0001` (1). Leading 1 → negative; magnitude = invert+1 of `1111 0001` = `0000 1111` = 15, so result = **−15** (1).

Examiner tip: −40 + 25 = −15. The sum `1111 0001` is negative — decode to confirm −15.

Q4 [3] (AO2). - (a) Normalisation gives the **greatest possible precision/accuracy** for a given number of mantissa bits (and a unique representation) (1). - (b) A + A·B = A·(1 + B) = A·1 = **A** (1 for answer A); law: **Absorption** (accept Distributive + identity) (1).

Examiner tip: absorption law A + A·B = A — the B term is redundant.

Q5 [4] (AO1). - (a) **Circuit switching** sets up a **dedicated end-to-end path** held for the whole communication; **packet switching** splits data into **packets routed independently**, sharing the network, reassembled at the destination (1 each, up to 2). - (b) Any two (1 each): **destination IP/address**, **source IP/address**, **packet/sequence number**, **TTL/hop count**, **protocol**.

Examiner tip: sequence number is needed because packets can arrive out of order and must be reassembled.

Q6 [3] (AO1/AO3). - Ethical issue (1–2): e.g. how the car should **prioritise lives** (passengers vs pedestrians), fairness of programmed decisions, transparency/accountability of the algorithm. - Legal responsibility (1): could be the **manufacturer/software developer**, the **owner/driver**, or shared — credit a reasoned choice (e.g. manufacturer if the autonomous system was in control). - Up to 3 total with reasoning.

Examiner tip: examiners want a reasoned position on responsibility, not just "it's complicated".

Q7 [4] (AO2). - (a) A **function returns a value**; a **procedure performs actions and need not return a value** (1). - (b) **By value**: a **copy** of the data is passed, so changes inside the subprogram **do not affect** the original (1). **By reference**: the **address/reference** is passed, so changes **do affect** the original variable (1). - (c) Local variables (1): avoid unintended side effects / name clashes; memory is **freed when the subprogram ends**; improves modularity and maintainability (any one).

Examiner tip: by value = copy (safe, isolated); by reference = original can be changed.

Q8 [5] (AO2). Bubble sort on `[7, 2, 9, 4, 1]`:

Pass	Result after pass
1	`[2, 7, 4, 1, 9]`
2	`[2, 4, 1, 7, 9]`
3	`[2, 1, 4, 7, 9]`
4	`[1, 2, 4, 7, 9]` ✓

- Correct list after pass 1 → `[2, 7, 4, 1, 9]` (1).
- Correct list after pass 2 → `[2, 4, 1, 7, 9]` (1).
- Correct lists after passes 3 and 4 → sorted `[1, 2, 4, 7, 9]` (1).
- **4 passes** needed before the list is sorted (1).
- Worst-case time complexity **$O(n^2)$** (1).

Examiner tip: one pass = comparing/swapping adjacent pairs left to right once. The largest value "bubbles" to the end each pass.

Q9 [4] (AO1/AO2). - (a) Insertion sort **$O(n^2)$** (1); merge sort **$O(n \log n)$** (1); binary search **$O(\log n)$** (1). - (b) For large n, **n log n grows much more slowly than n^2**, so merge sort scales far better / does far fewer operations (1).

Examiner tip: $O(n \log n)$ beats $O(n^2)$ as n grows — the gap widens rapidly for large datasets.

Q10 [6] (AO3). Dijkstra's algorithm, A → E. Tentative distances from A ("—" = settled):

Step	Current	A	B	C	D	E	Visited
Init	–	0	∞	∞	∞	∞	{}
1	**A** (0)	—	**2**	5	∞	∞	{A}
2	**B** (2)	—	—	min(5, 2+1)=**3**	min(∞, 2+7)=9	∞	{A,B}
3	**C** (3)	—	—	—	min(9, 3+3)=**6**	min(∞, 3+8)=11	{A,B,C}
4	**D** (6)	—	—	—	—	min(11, 6+2)=**8**	{A,B,C,D}
5	**E** (8)	—	—	—	—	—	{A,B,C,D,E}

- Correct initialisation: A = 0, others ∞ (1).
- B relaxed to 2, then C relaxed to 3 via B (1).
- Visits nodes in correct distance order **A, B, C, D, E** (1).
- Correct relaxation of D to 6 via C, and E updated to 8 via D (1).
- Shortest distance **A → E = 8** (1).
- Shortest path **A → B → C → D → E** (2 + 1 + 3 + 2 = 8), by tracing predecessors E ← D ← C ← B ← A (1).

Examiner tip: always settle the smallest unvisited tentative distance next. Note A→C direct (5) is beaten by A→B→C (3) — relax carefully. State both distance (8) and path.

Q11 [6] (AO3). Levelled extended response (up to 6). Indicative content across the three strands: - **(i) Abstraction & decomposition**: decompose into modules — record storage, search, reminders, networking/security — built and tested separately; abstraction hides detail (e.g. model an appointment as date/time/patient ID, ignore irrelevant detail) so complexity is manageable (up to 2). - **(ii) Data structure / search**: justify a **sorted structure + binary search O(log n)** or a **hash table O(1) average** for fast lookup, contrasted with linear search O(n) which is too slow for many records; reference time complexity explicitly (up to 2). - **(iii) Legal**: patient data is **sensitive personal data** under data protection law (GDPR / Data Protection Act) — needs a lawful basis, **secure storage/encryption**, access control, and retention limits (up to 1–2). - Mark holistically: a strong answer addresses all three strands with applied, justified reasoning; weaker answers describe one strand only.

Examiner tip: this is a synoptic question — explicitly tie computational-thinking choices to a complexity justification AND name the relevant legislation for full credit.

Total: 46 marks.